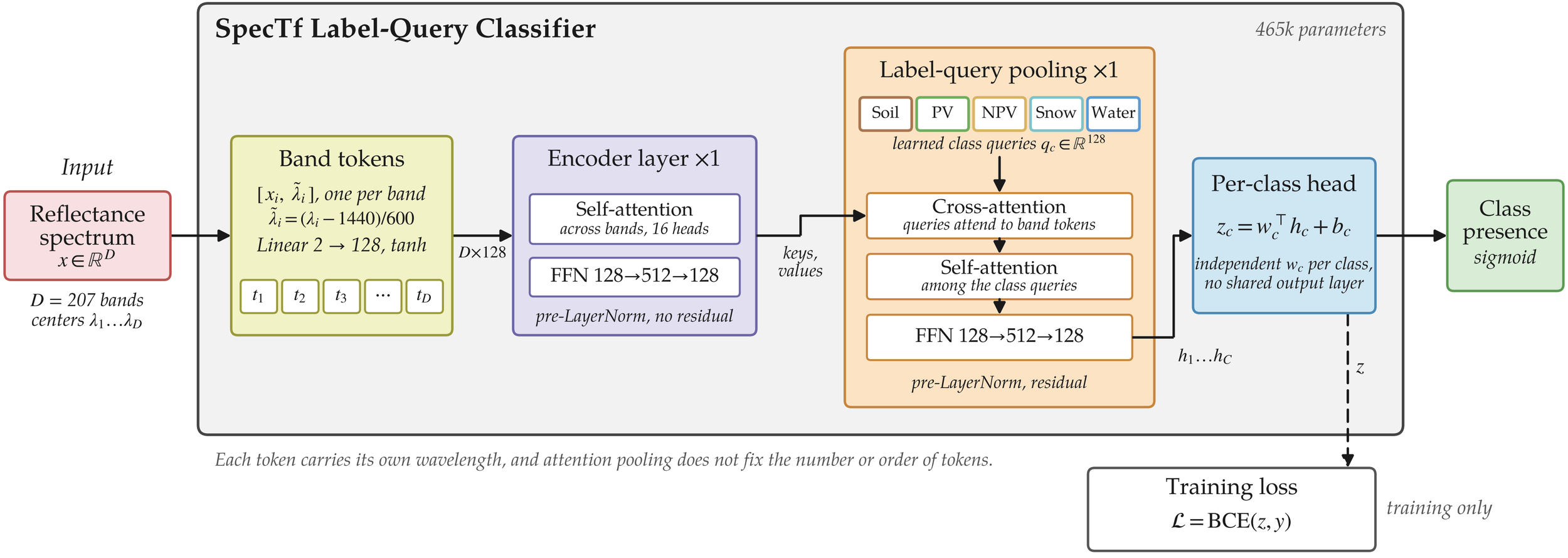

SpecTf Label-Query Classifier
465k parameters
Input
Reflectance spectrum
$x \in \mathbb{R}^D$
$D = 207$ bands
centers $\lambda_1 \ldots \lambda_D$
Band tokens
$[x_i, \tilde{\lambda}_i]$, one per band
$\tilde{\lambda}_i = (\lambda_i - 1440)/600$
Linear 2 → 128, tanh
t_1 t_2 t_3 ⋯ t_D
D×128
Encoder layer ×1
Self-attention
across bands, 16 heads
FFN 128→512→128
pre-LayerNorm, no residual
keys, values
Label-query pooling ×1
Soil PV NPV Snow Water
learned class queries $q_c \in \mathbb{R}^{128}$
Cross-attention
queries attend to band tokens
Self-attention
among the class queries
FFN 128→512→128
pre-LayerNorm, residual
$h_1 \ldots h_C$
Per-class head
$z_c = w_c^\top h_c + b_c$
independent w_c per class,
no shared output layer
z
Class presence
sigmoid
Each token carries its own wavelength, and attention pooling does not fix the number or order of tokens.
Training loss
$\mathcal{L} = \mathrm{BCE}(z, y)$
training only